

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
 B. Sc. Engineering 1st Year 2nd Term Regular Examination, 2023
CE 1201
 (Engineering Mechanics)

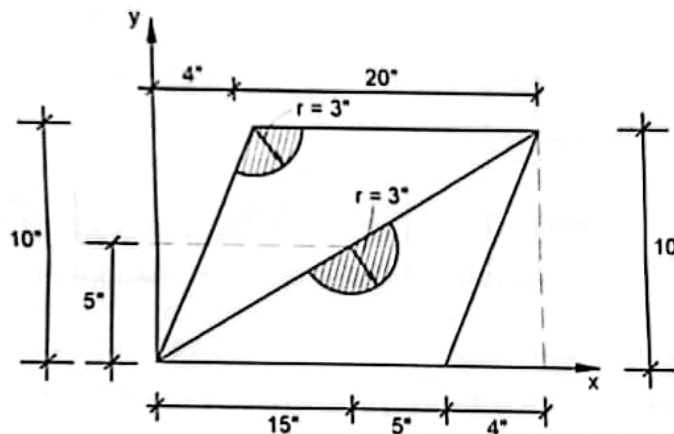
Full Marks: 210

Time: 3 hrs

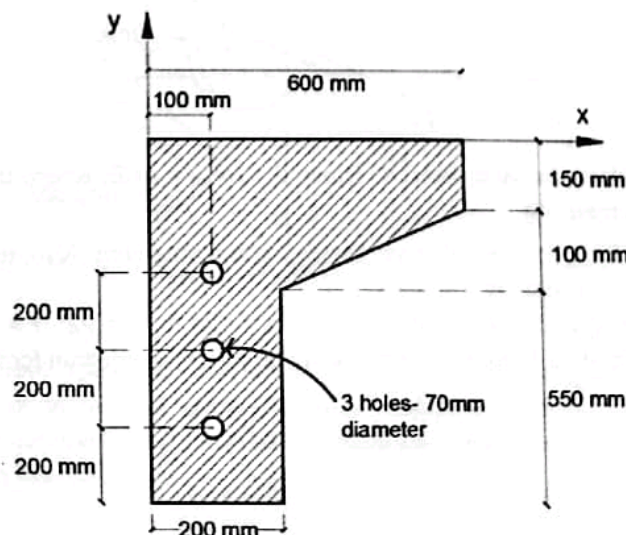
- N.B. i) Answer any three questions from each section in separate script.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable value for any missing data.

Section – A

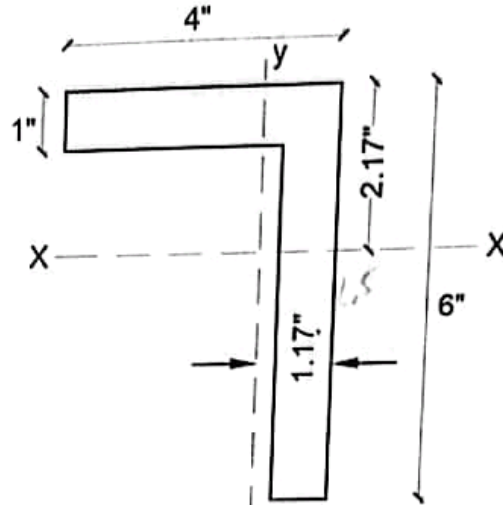
1. (a) Distinguish between centroid and center of gravity. What are the real-life applications of centroid in the field of building engineering? (07)
- (b) Locate the centroid of the section of a circle of radius r , whose central angle is 2θ . (10)
- (c) Find the centroid about X-axis and Y-axis of the unshaded area. (18)



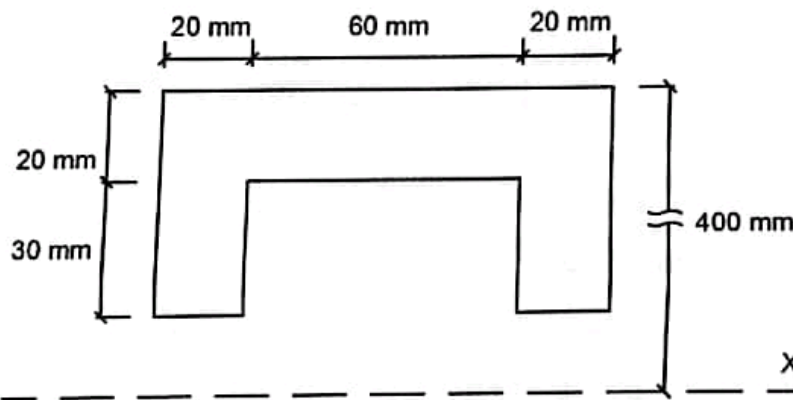
2. (a) Define moment of inertia and product of inertia. Find the moment of inertia and radius of gyration of a triangular area about a base. (12)
- (b) Calculate the polar moment of inertia and radius of gyration of the cross-section about parallel axis and centroidal axis. (23)



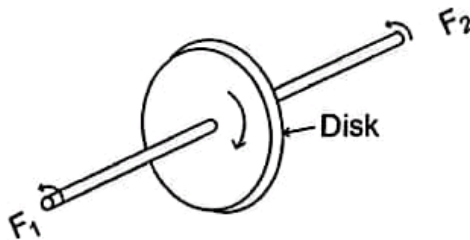
3. (a) An angle section is shown in the following figure for which $I_x = 30.8 \text{ in}^4$, $I_y = 10.8 \text{ in}^4$. (12)
Determine the maximum and minimum centroidal moment of inertia and the radius of gyration. Also, compute the product of inertia.



- (b) The outside diameter of a pulley is 0.80m and the cross section of its rim is shown in figure. Knowing that the pulley is made of steel, determine the mass and weight of the rim by using Pappus-Guldinus theorem. Here, $\rho = 7.85 \times 10^3 \text{ kg/m}^3$. (10)



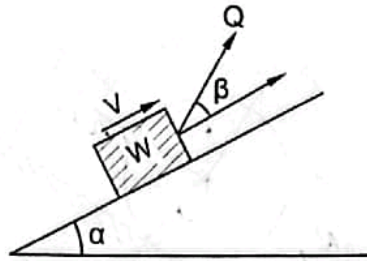
- (c) A 644 lb disk, 6 ft in diameter, is keyed to a 322 lb shaft which is 6 inch in diameter. (13)
The shaft turning at 720 rpm, is supported by two bearings, where $f = 0.02$. If there is no torque on the rotating members except that due to the frictional force in the bearings, how long does it take for the angular speed to be reduced to 120 rpm?



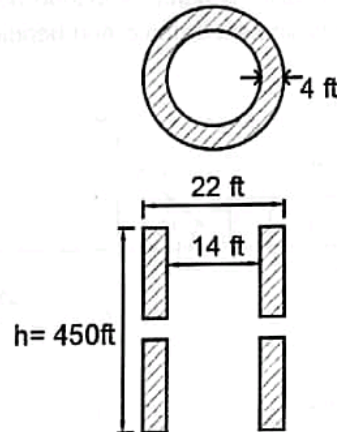
- (b) What is classical dynamics? Prove that $W_{\text{net}} = \Delta KE$, where the symbols bear their usual meaning. (10)

- (b) State the principles of linear impulse and momentum. Also, explain the terms work and kinetic energy. (15)

In the figure, $W = 109 \text{ lb}$, $\alpha = 39^\circ$, $Q = 129 \text{ lb}$, $\beta = 22^\circ$, $f = 0.25$ and the body is displaced 12 ft up the inclined surface. Find the resultant force on W and net work done.

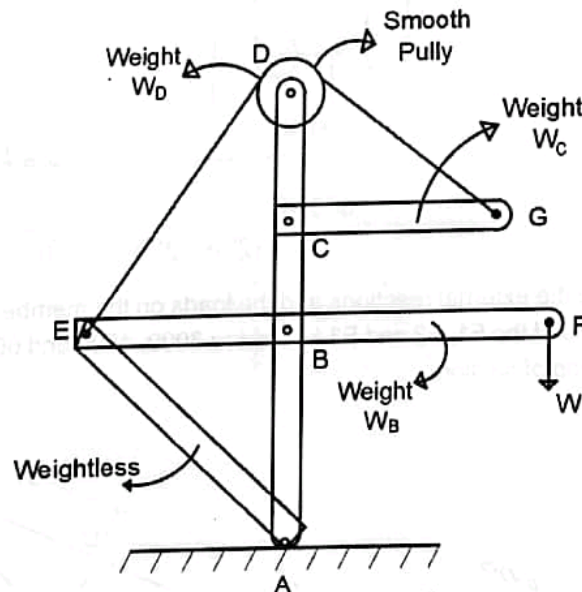


- (a) What is work done against gravity in lifting the material to its place during the build of a masonry chimney which is 450 ft high, 22 ft outer diameter and 14 ft inner diameter? The masonry weight 100 lb/ft³. (10)

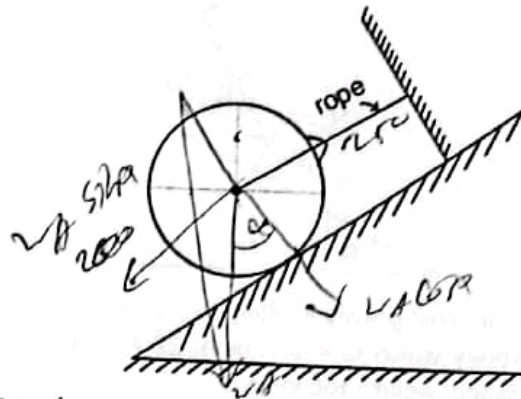


Section - B

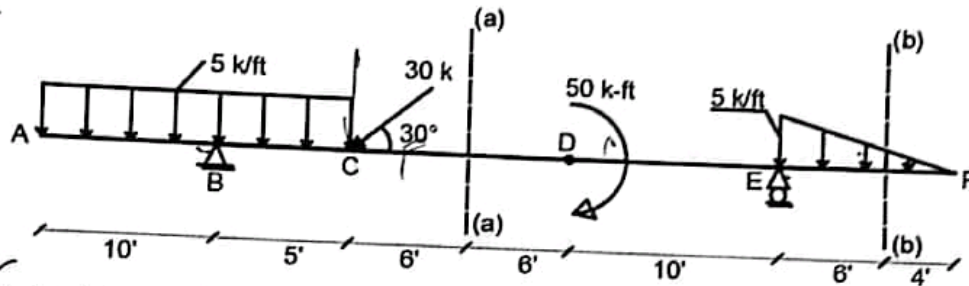
5. Define the following term: (i) Transmissibility of force, (ii) Angle of Repose, (iii) Wedge and (iv) Co-planar force system. (20)
- Draw the free-body diagram of the following body shown in the figure below.



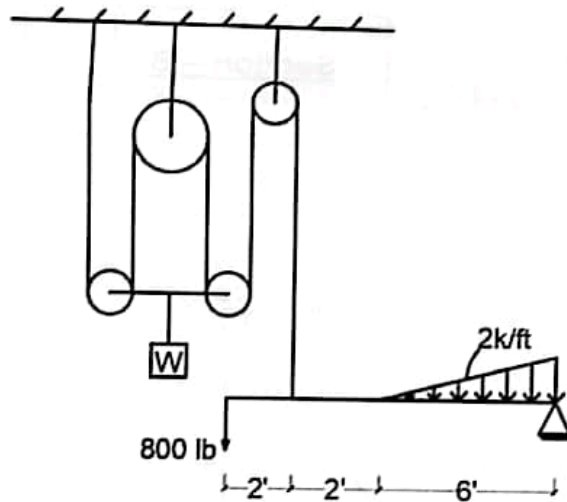
- (b) Illustrate different types of support systems. A wheel shown in figure given below is on the point of equilibrium using a tension rope placed parallel on the inclined surface. If the tension force on the rope is 250 lb then find the inclination of the plane and the plane reaction. (15)



Define torque and couple. A beam is loaded as shown in figure below. Find the reactions at supports and shear force and bending moment at section (a)-(a) and (b)-(b). (20)



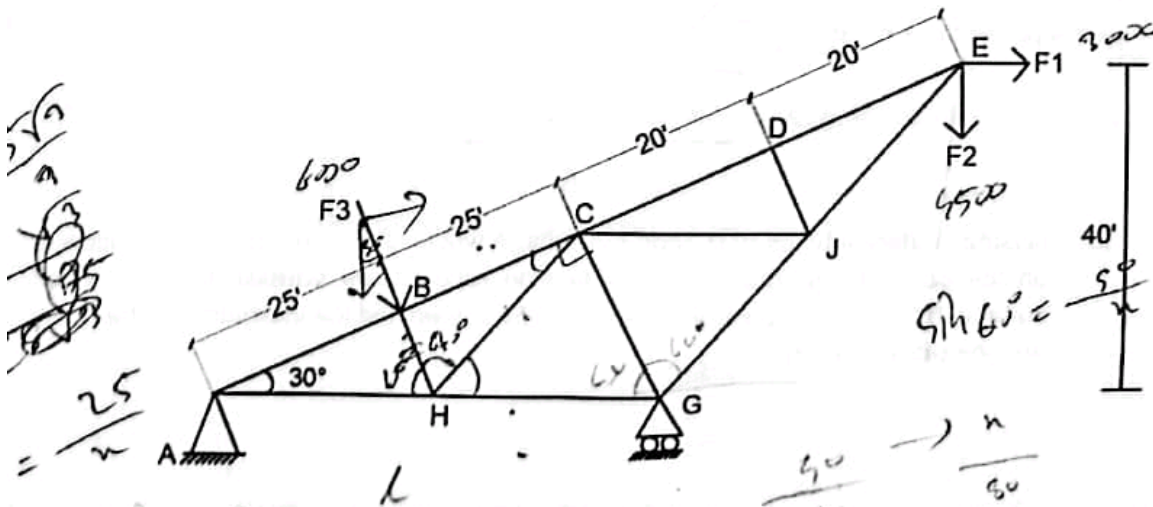
In the following figure what will be the value of 'W' if equilibrium exists? Assume the pulleys are frictionless and weightless. (15)



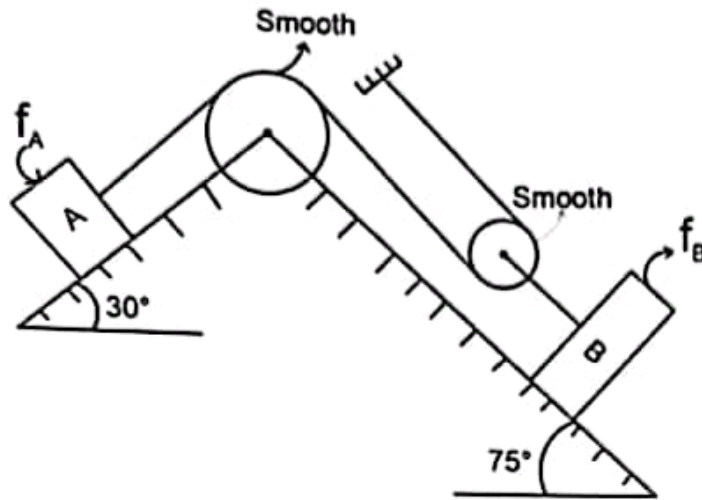
Full



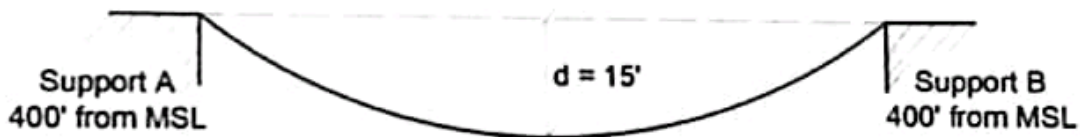
Compute the external reactions and the loads on the members GH, CG and GJ. The magnitude of the F_1 , F_2 and F_3 forces are 3000, 4500 and 6000 lb respectively. Use the method of section. (20)



- ✓ In the following figure, weight of body A is 400 lb, $f_A = 1/3$ and $f_B = 1/4$. If the body B is on the point of moving downward, determine the tension in the cable and weight of body B. (15)



- (a) What is parabolic chord? For a typical parabolic chord derive that, $y = \frac{wx^2}{2Q}$ (08)
- (b) A 60 ft cable shown in figure below weights 360 lb. For the support condition illustrated in the figure, determine the tension of the support and the span. (12)



- (c) Find the force Q on the top of the wedge. Given that, $W_A = 200$ lb, $W_B = 500$ lb, $W_C = 5000$ lb and $R = 8000$ lb. Consider, $f = 1/3$ for all the slippery surface, what force Q will impose impending motion of C. (15)

